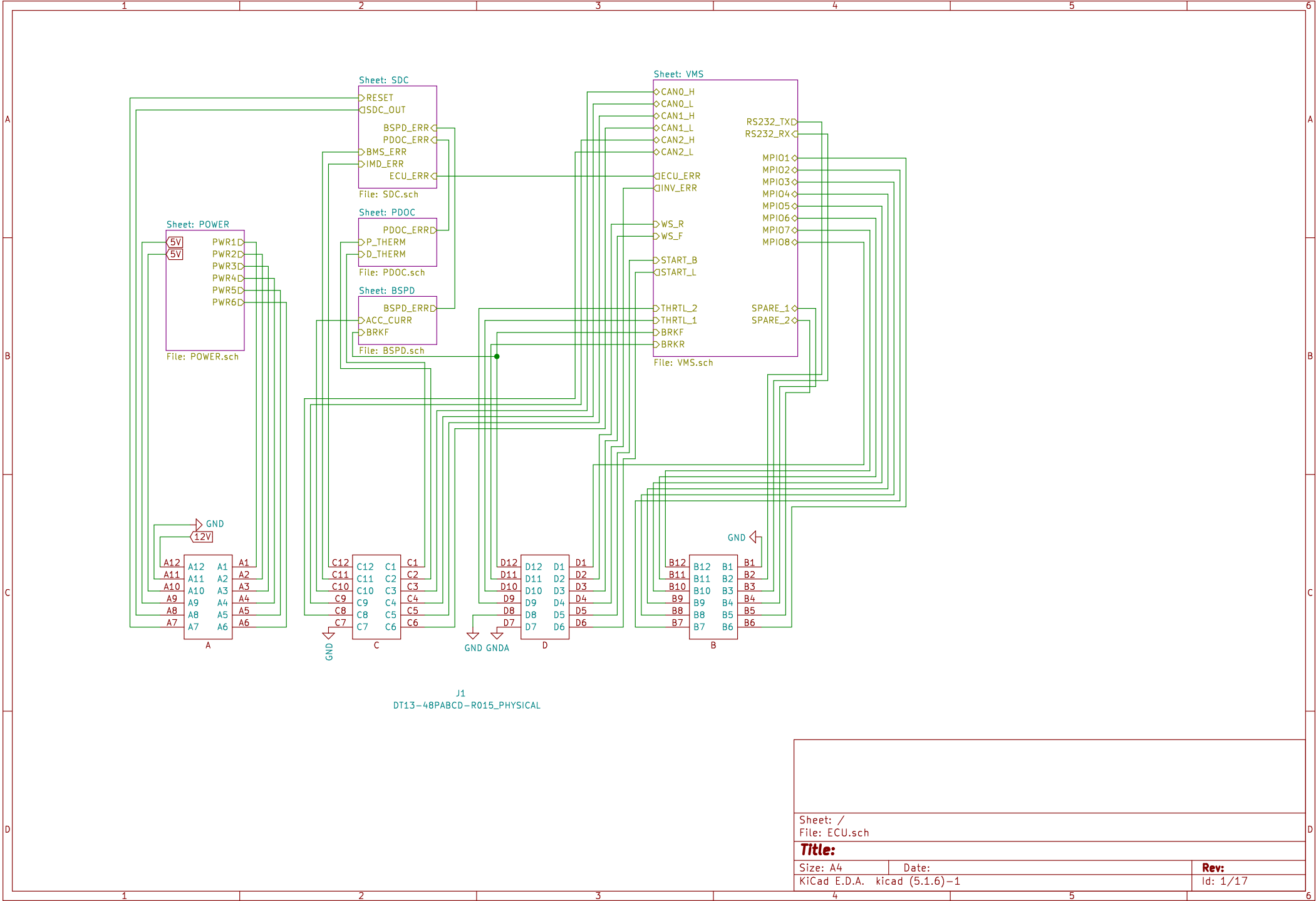
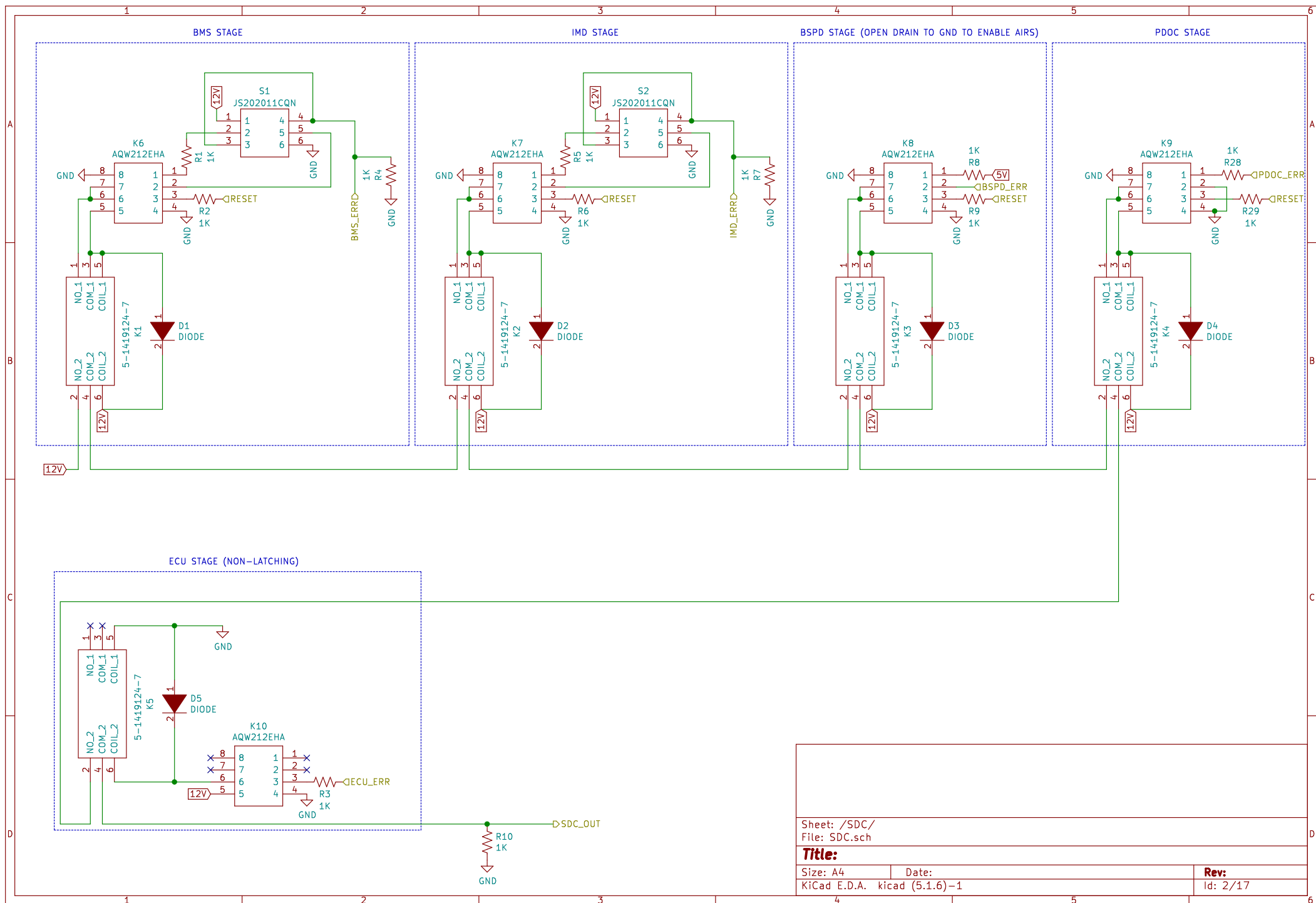




Sheet: /		
File: ECU.sch		
Title:		
Size: A4	Date:	Rev:
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Sheet: /SDC/
File: SDC.sch

Title:

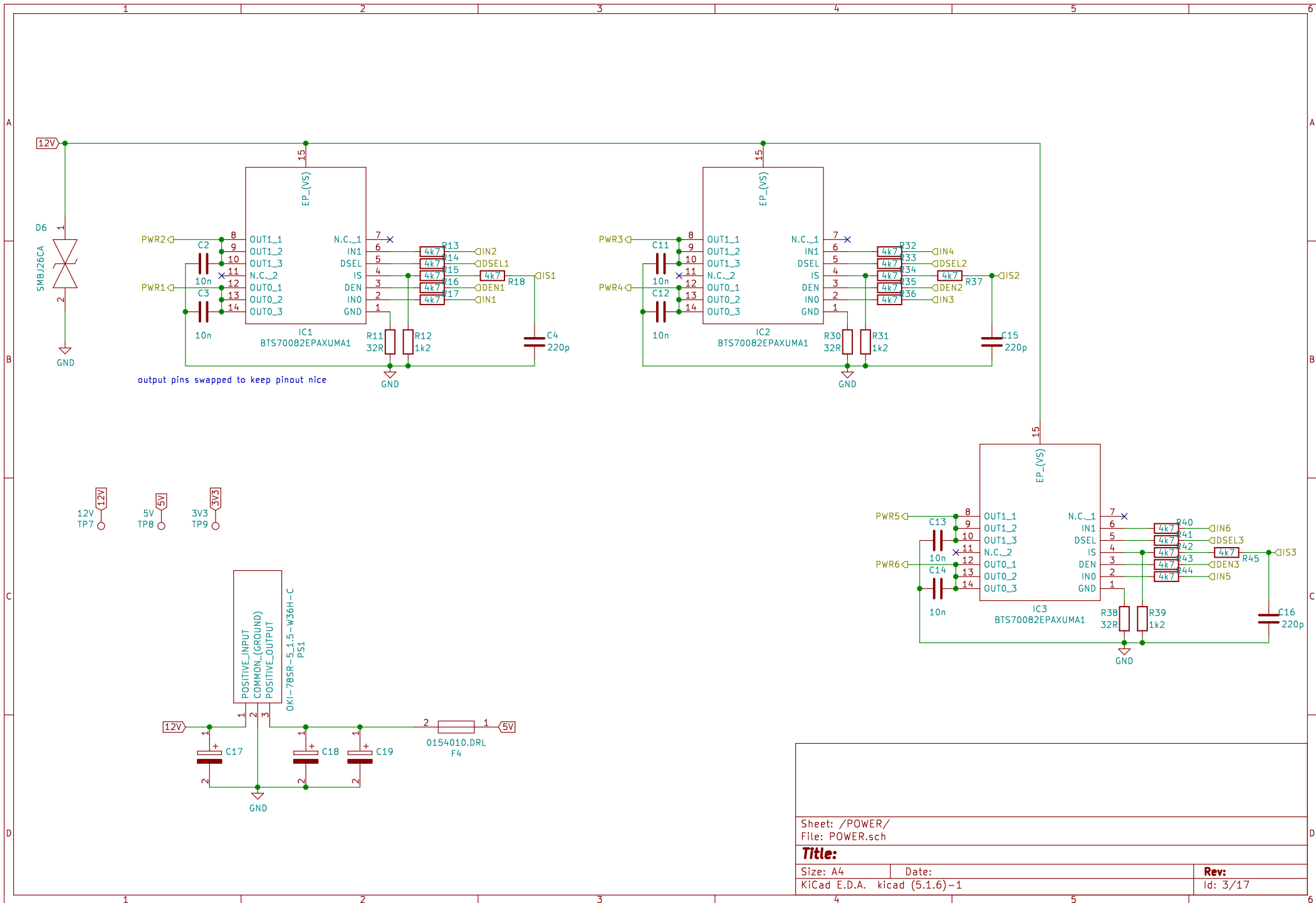
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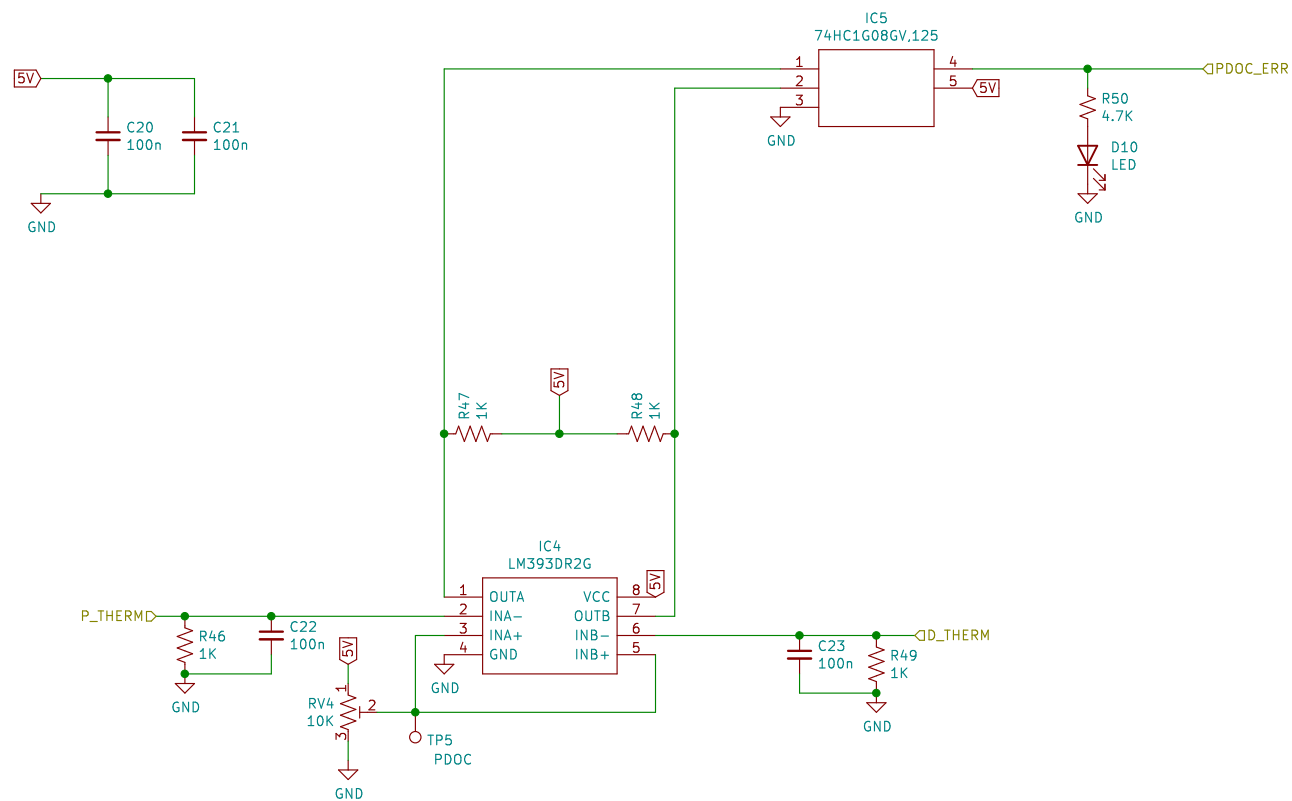
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Sheet: /PDOC/
File: PDOC.sch

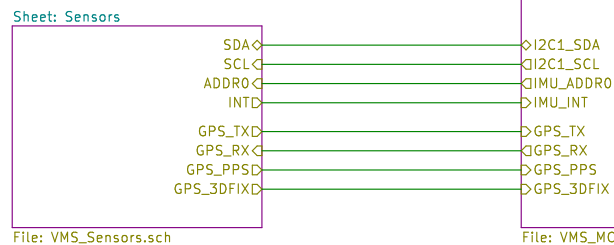
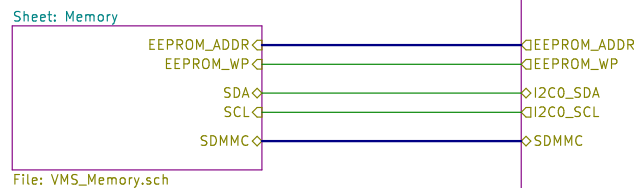
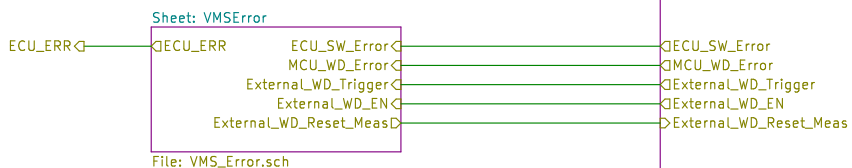
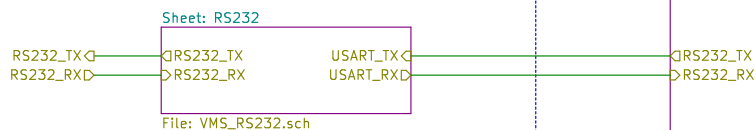
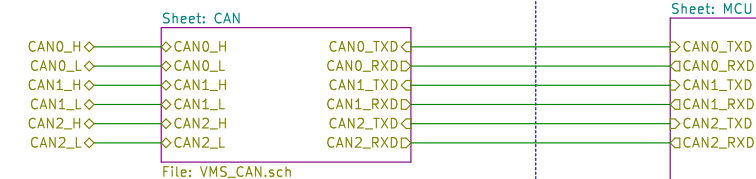
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Serial Comms

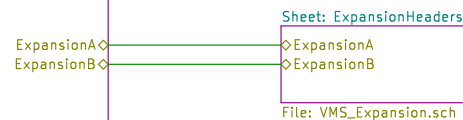
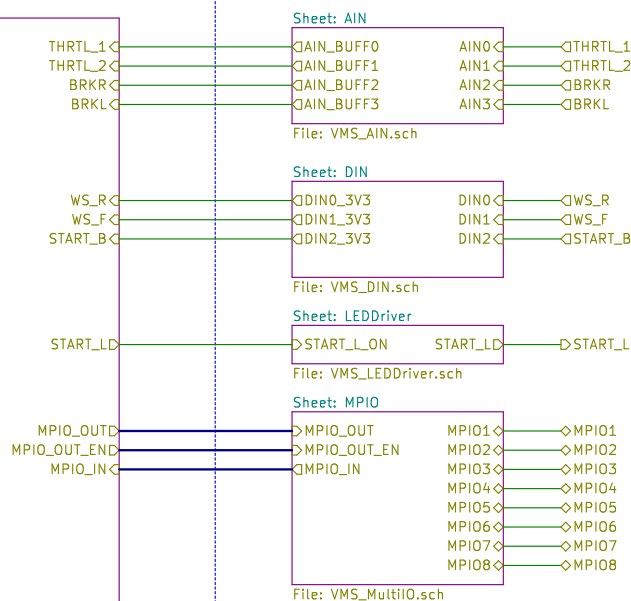


Sheet: MCU

File: VMS_MCU.sch

All signals in/out of MCU are 3.3V

Discrete Signals



TODO:

- * Tie 5V in power section to 5V power label
- * Create 3v3 regulator
- * Internal expansion headers
- * Power LEDs
- * Blinker LED
- * VDDA analog reference

Sheet: /VMS/
File: VMS.sch

Title:

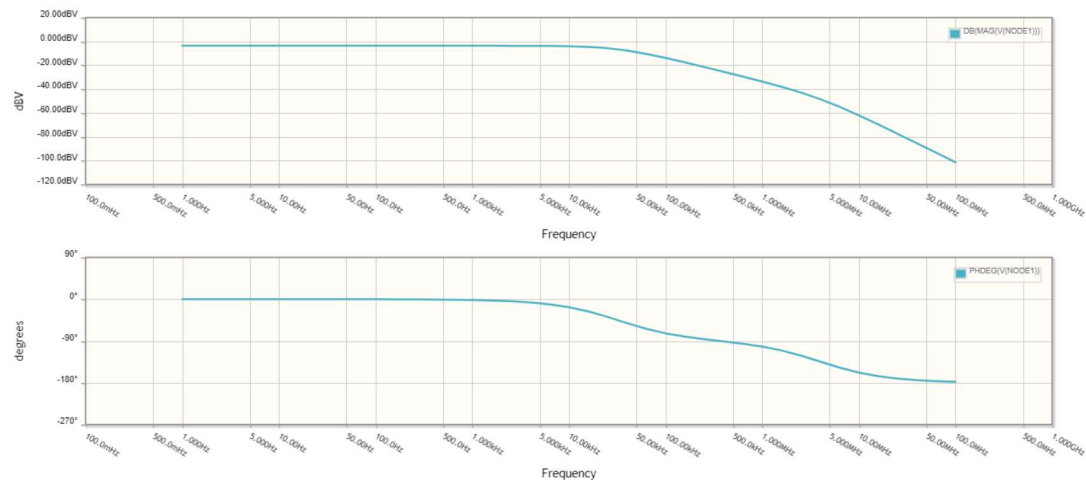
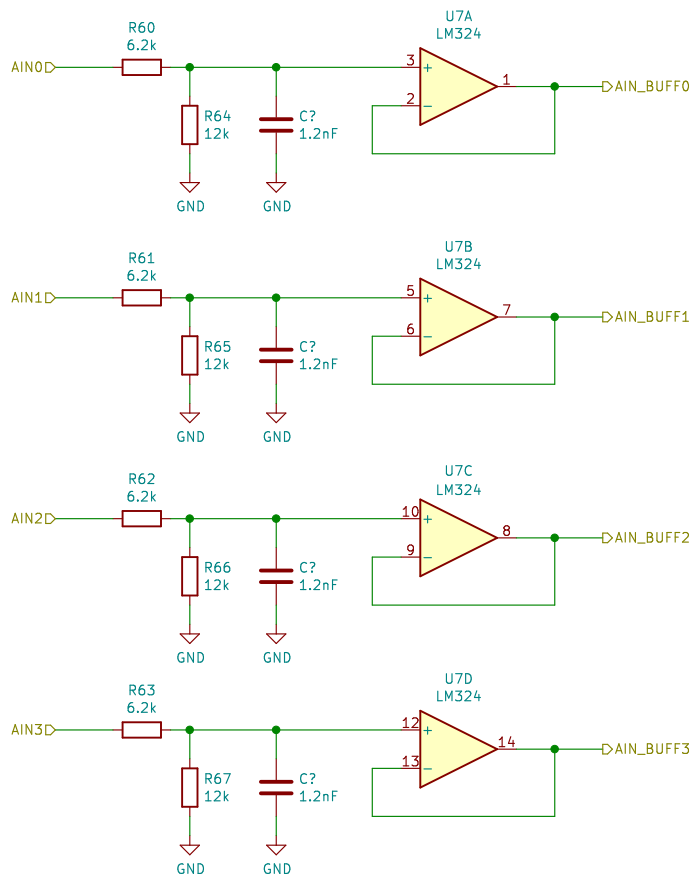
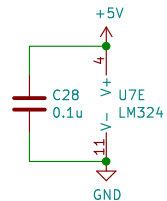
Size: A4

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0-5V to 0-3.3V (0.66G ideally) and pulled down on input
 6.2k & 12k => 0.659G (5V -> 3.297V)
 1.2nF gives F_c at 20kHz

Sheet: /VMS/AIN/
 File: VMS_AIN.sch

Title:

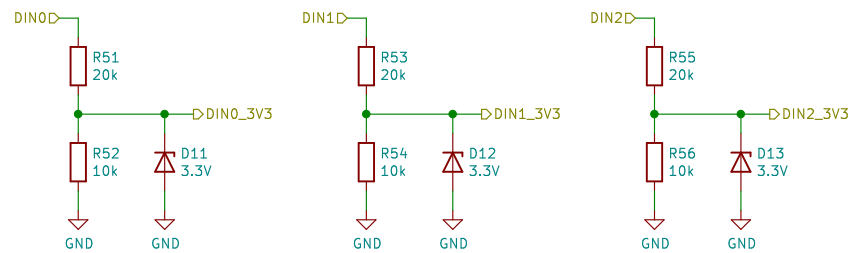
Size: A4

Date:

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Sheet: /VMS/DIN/
File: VMS_DIN.sch

Title:

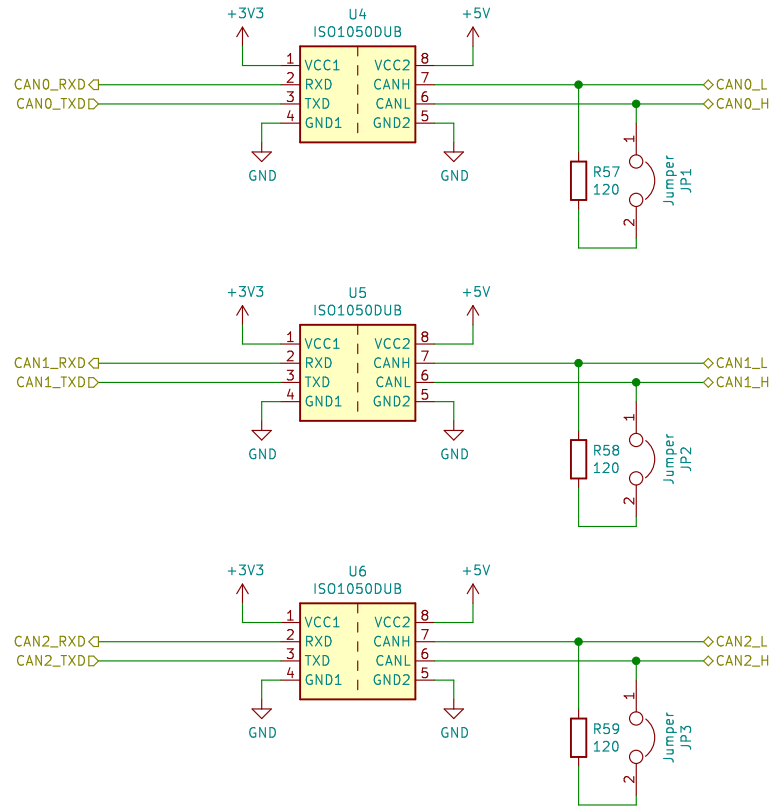
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Date:

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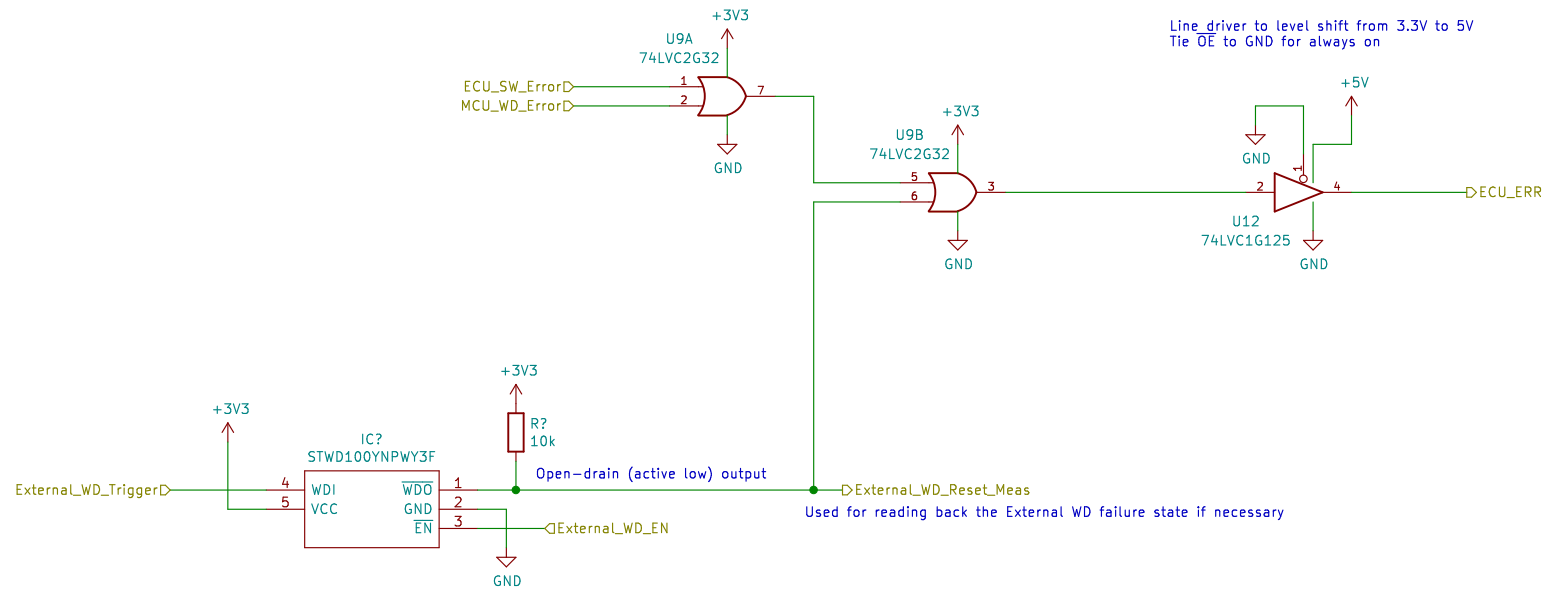
Rev:

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TODO
* Check exact ISO1050 unit. DUB?

Sheet: /VMS/CAN/ File: VMS_CAN.sch		
Title:		
Size: A4	Date:	Rev:
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TODO

- * The 74LVC schematics often have weird issues in tying VDD/VCC to +3V3
- * Address latching issue and logic inversion
- * The SDC can actually take 3.3V so don't need the output buffer

Sheet: /VMS/VMSError/
File: VMS_Error.sch

Title:

Size: A4

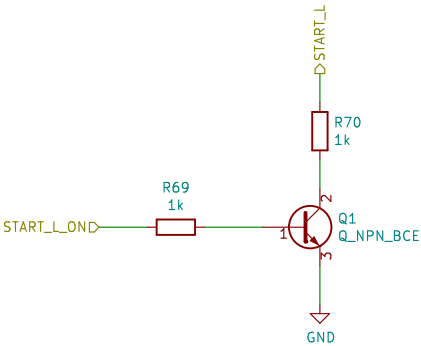
Date:

KiCad E.D.A. kicad (5.1.6)-1

Rev:

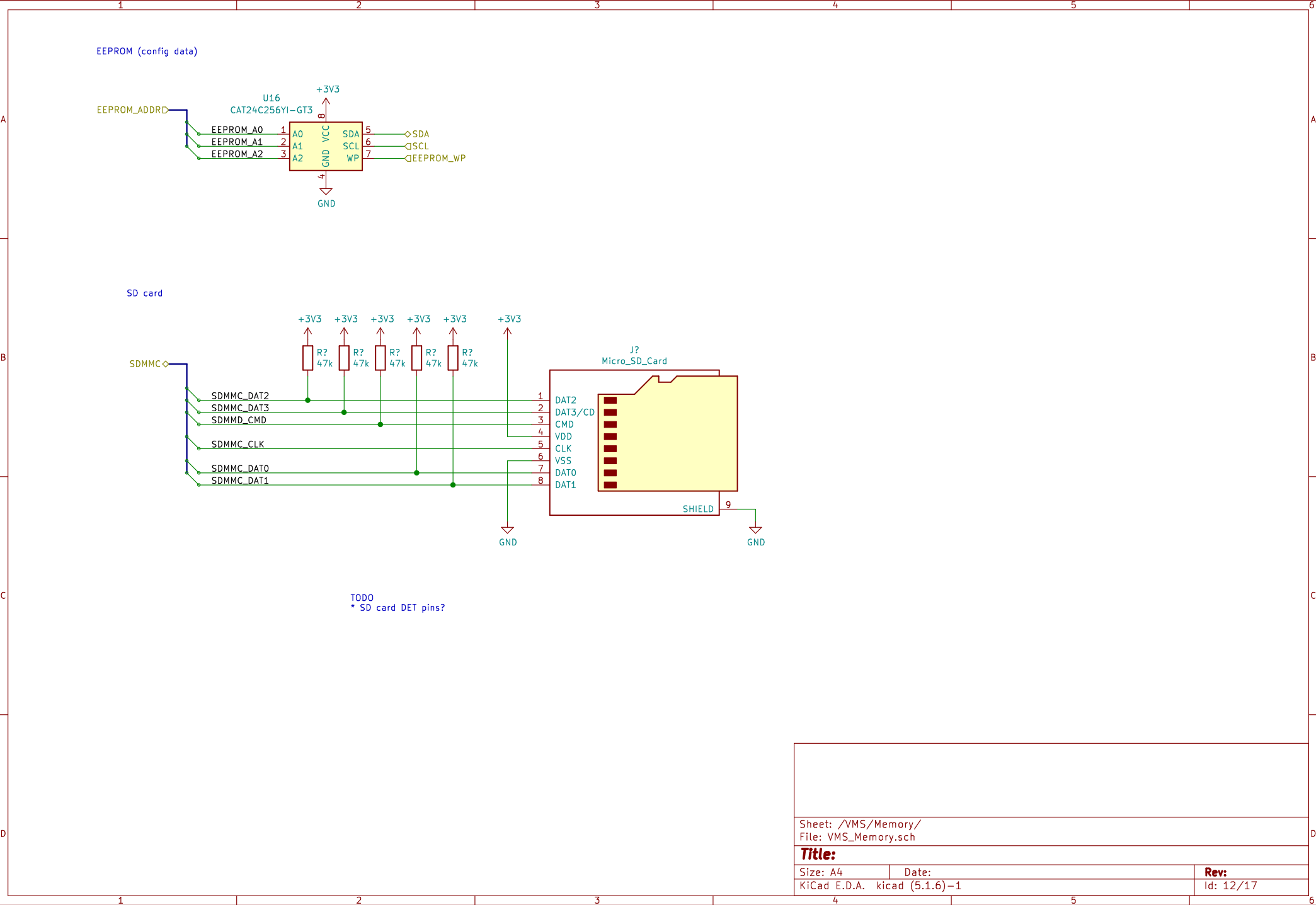
Id: 10/17

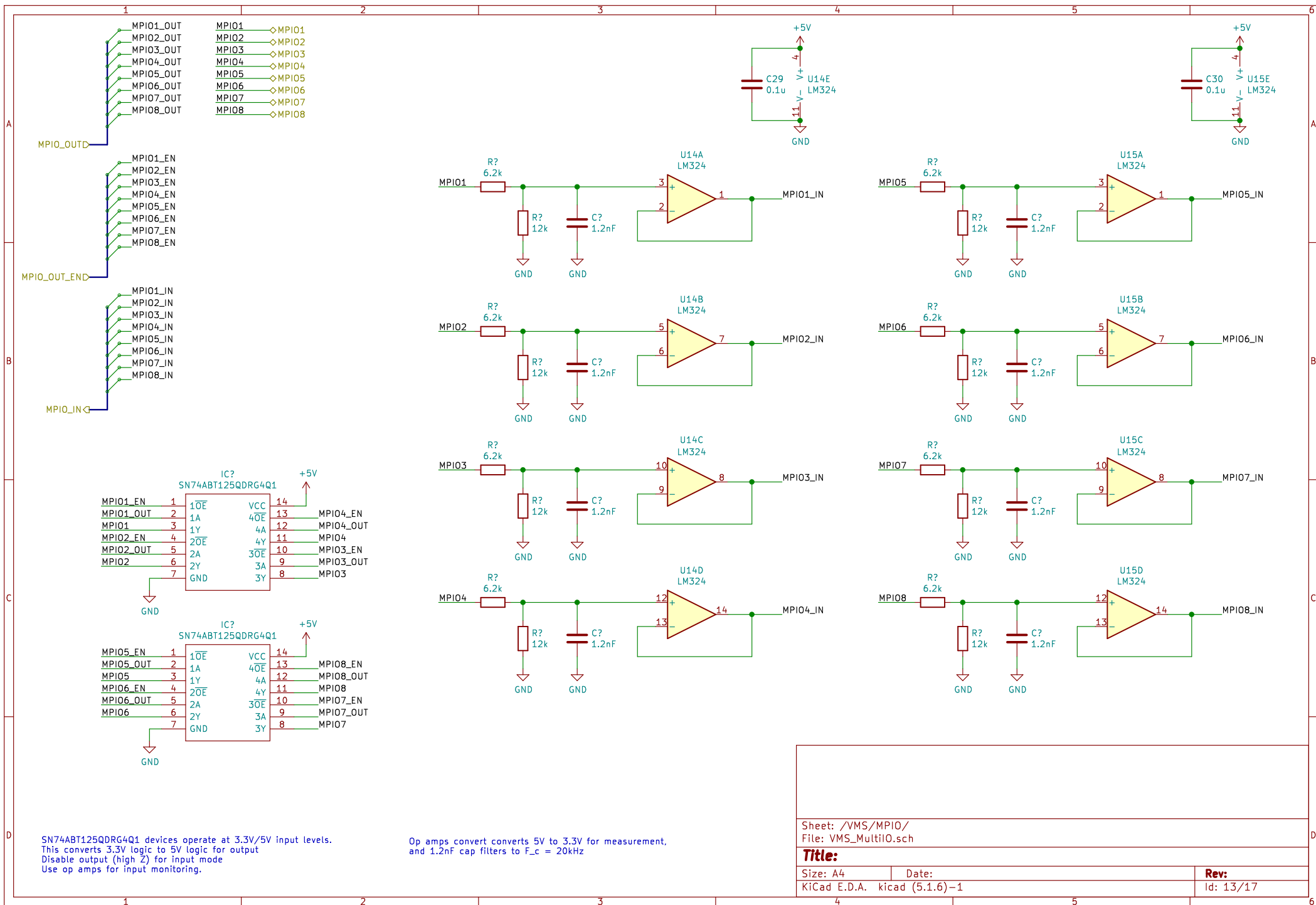
STARTL_ON is a 3V3 signal from MCU
STARTL is the output to be in series with the LED on dash



TODO
* Resistor sizing
* Pick BJT part number

Sheet: /VMS/LEDDriver/ File: VMS_LEDDriver.sch		
Title:		
Size: A4	Date:	Rev:
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SN74ABT125QDRG4Q1 devices operate at 3.3V/5V input levels.
This converts 3.3V logic to 5V logic for output
Disable output (high Z) for input mode
Use op amps for input monitoring.

Op amps convert 5V to 3.3V for measurement,
and 1.2nF cap filters to $f_c = 20\text{kHz}$

Sheet: /VMS/MPIO/
File: VMS_MultiIO.sch

Title:

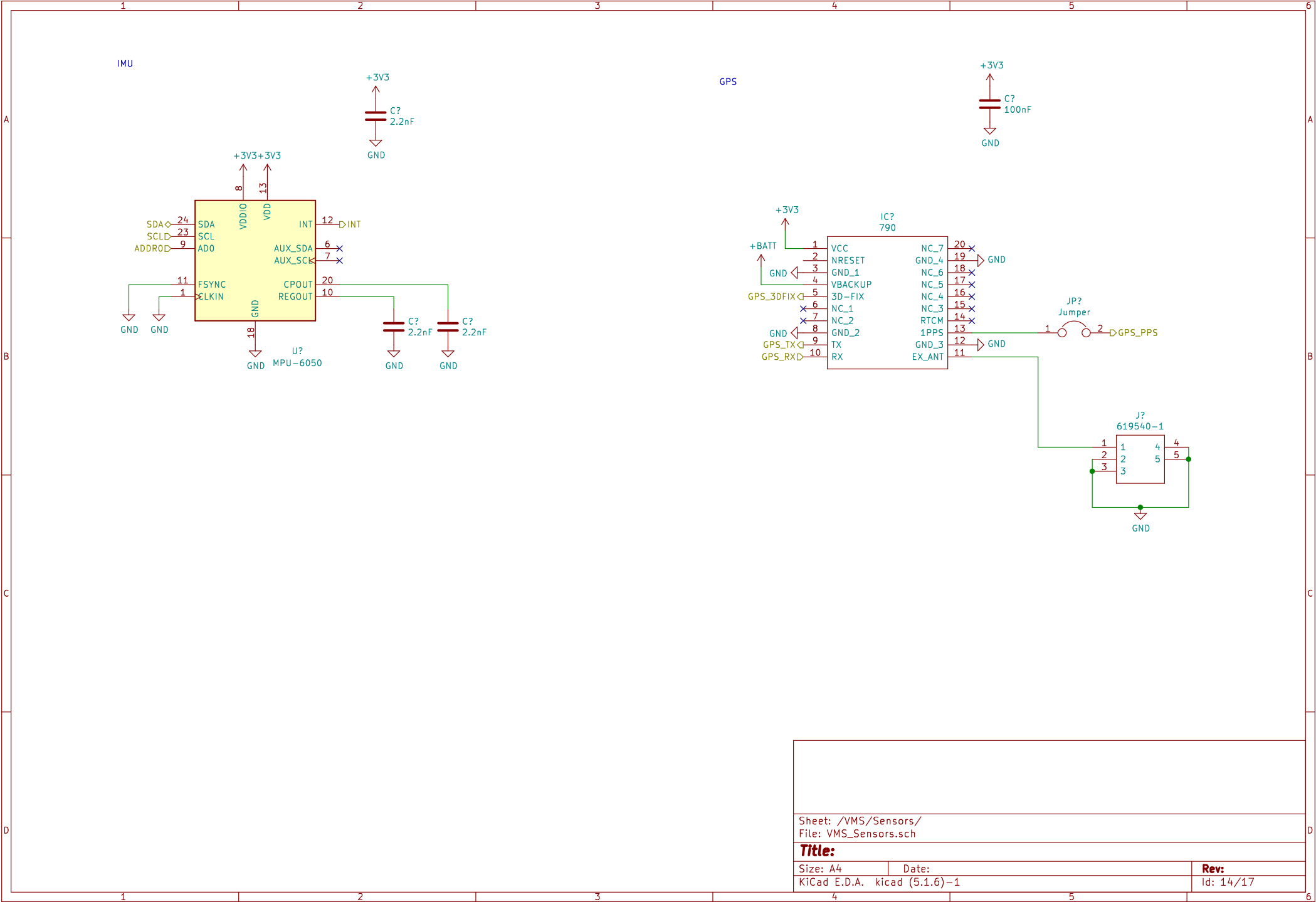
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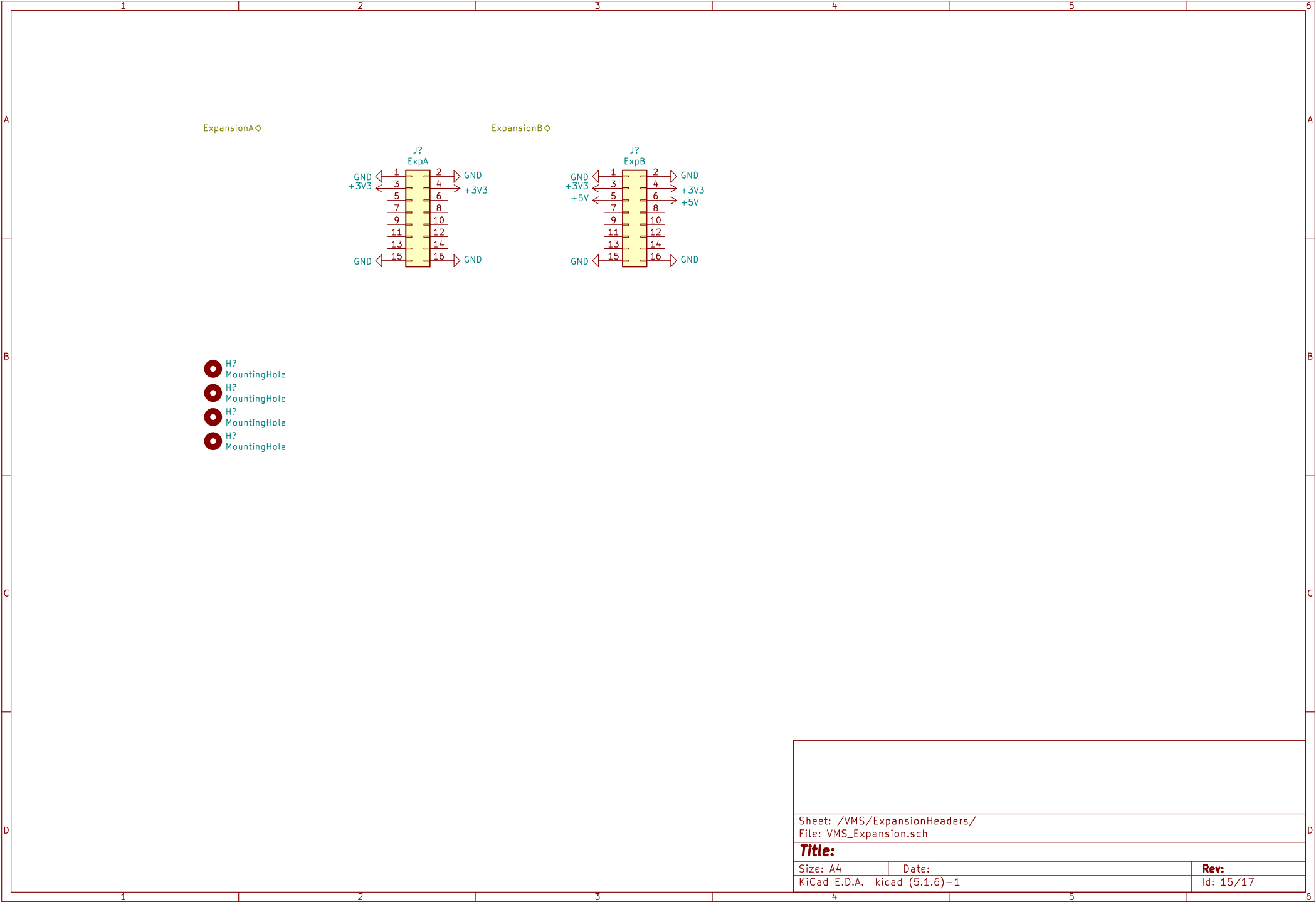
Date:

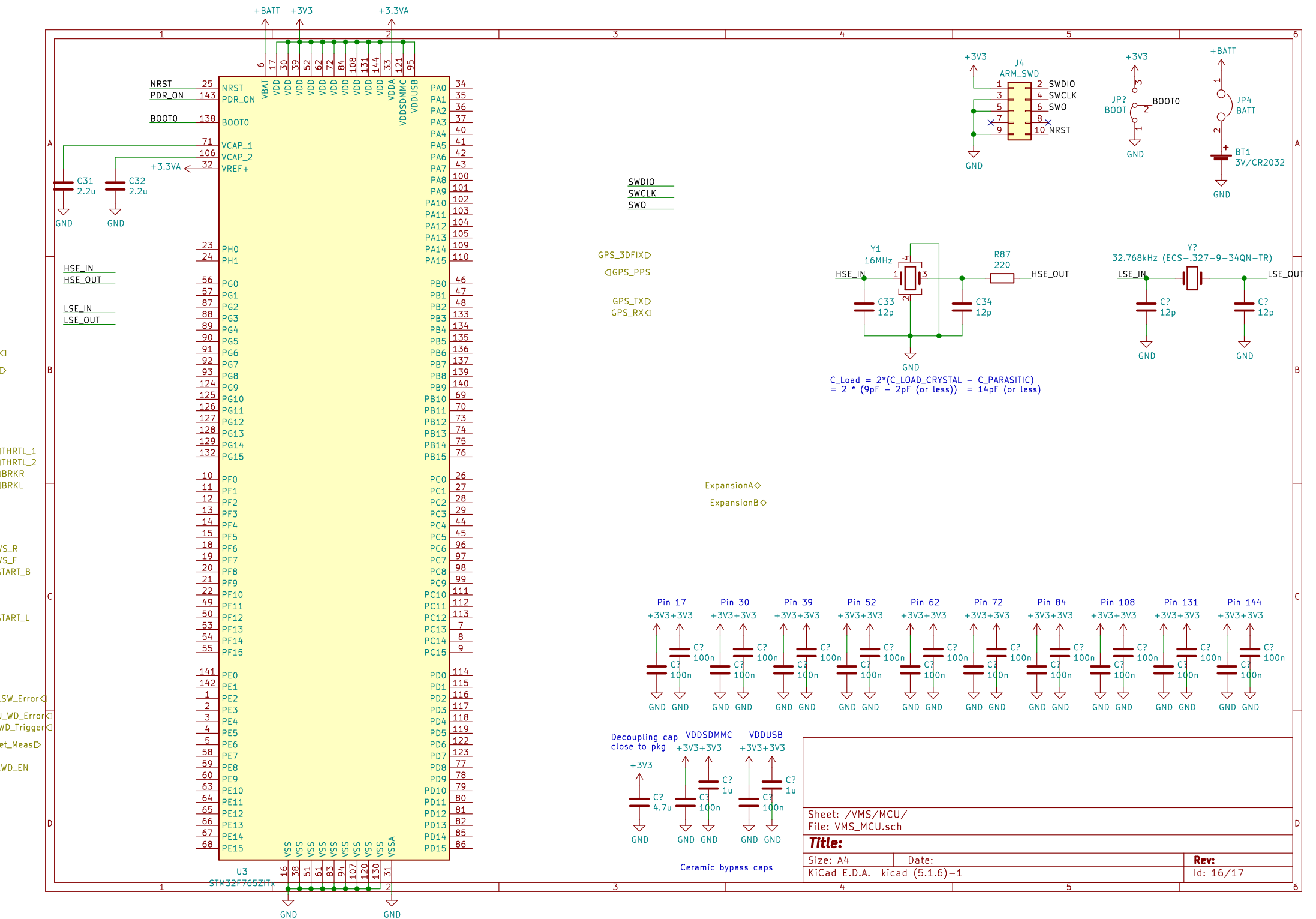
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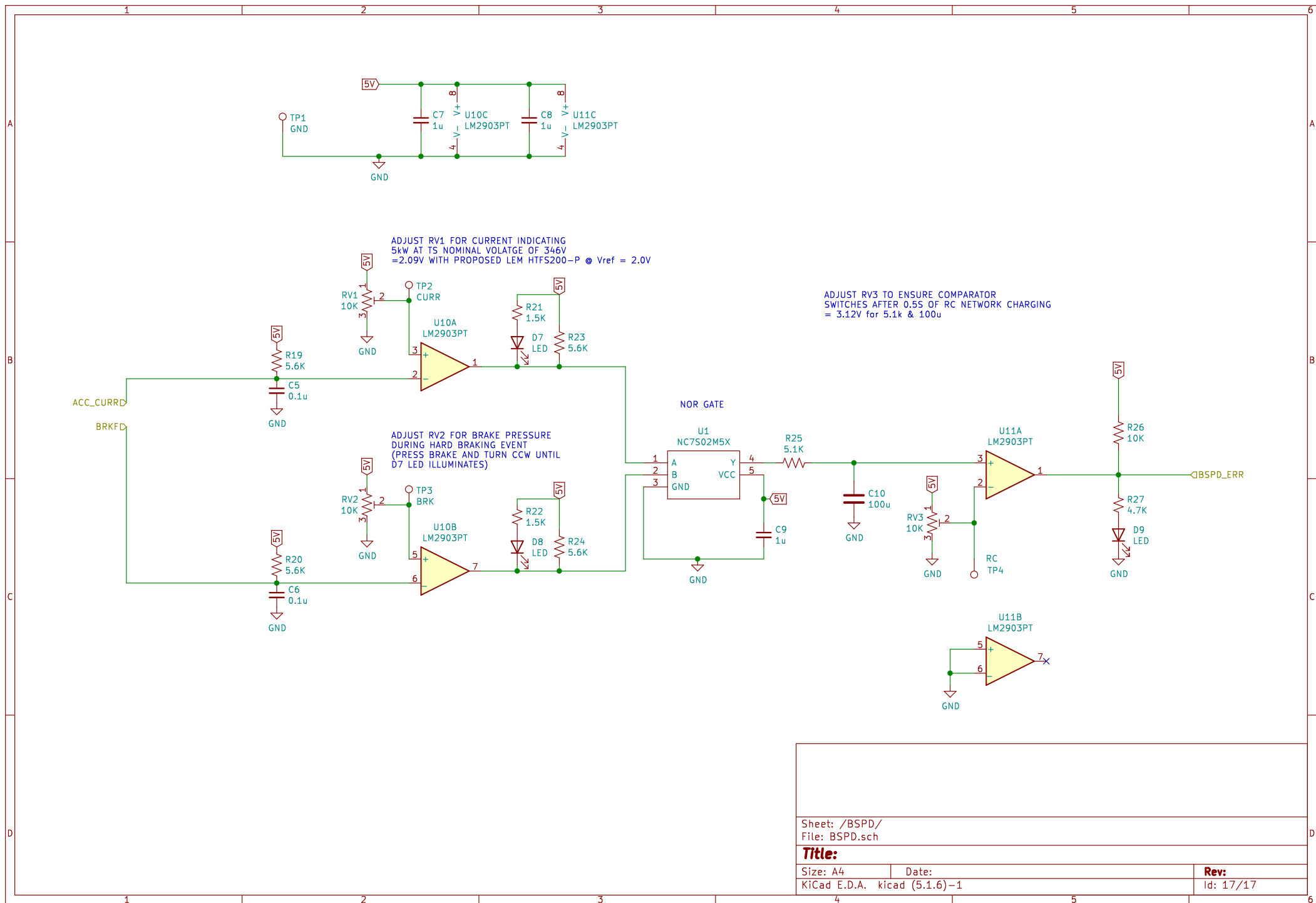
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Sheet: /BSPD/
File: BSPD.sch

Title:

Size: A4

Date:

KiCad E.D.A. kicad (5.1.6)-1

Rev:

Id: 17/17